As illustrated in Fig. 2.6, since k and $k-q$ are restricted to be close the two points at $\pm k_F$, there are only two possible value of q depending on the sign of k and $k-q$. When k and $k-q$ have the same sign, $|q| \sim 0$. This corresponds to a scattering term for fermions moving in the same direction and might be interpreted as a *forward scattering.* When k and $k-q$ have opposite sign, $|q| \sim 2k_F$, meaning that in this excitation fermions exchange side (*backward scattering*).

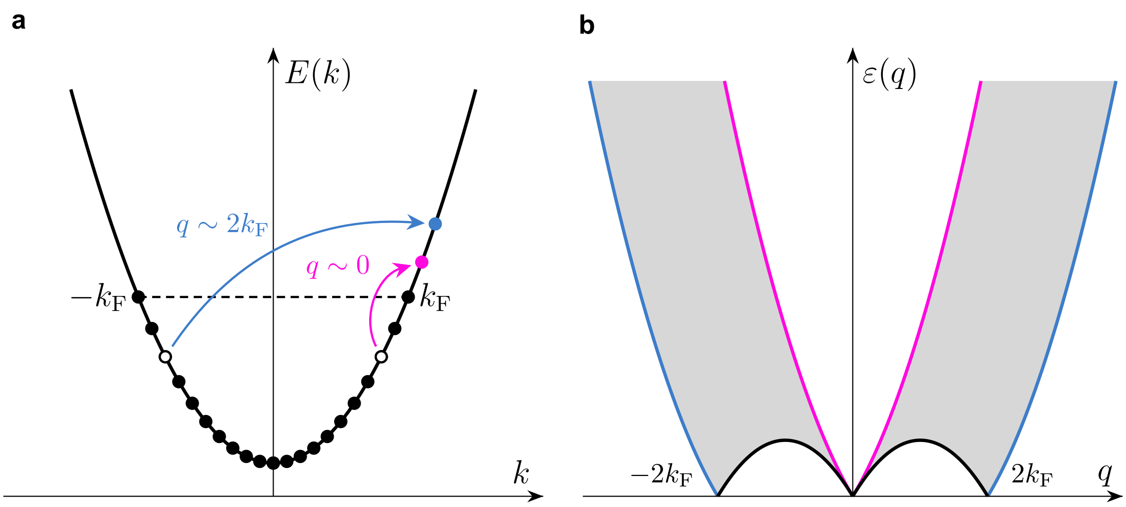

Figure 2.6: Excitation Spectrum for 1D fermions. **(a)** In one dimension, the Fermi surface constits of only two points, corresponding to wave vector $\pm k_F$. Low-energy single-particle excitations such asforward (pink) and backward (blue) scattering for small momentum . The $q \sim 0$ component switches on particle-hole excitations on the same branch, whereas the $q \sim 2k_F$ carries a particle across the two sides of the Fermi surface. Black dots are filled (occupied) states, holes are empty circles. **(b)** Particle–hole excitation spectrum. Accessible energies are the one shaded in grey, whereas the forbidden regions indicated by the white gaps are unique to the 1D nature of these systems and find no correspondece to higher dimensions. The position of the spectral edges, separating accessible from forbidden regions, are marked by a solid black curve. Experimental observations of the magnetic response for certain classes of solid-state materials (Heisenberg antiferromagnetic chains such as $KCuF_3$) through inelastic neutron scattering have confirmed these regions [164, 165]. Adapted from [166].

The problem with the term in Eq. 2.26 is being the product between two density operators, when these are expressed in the fermion operator basis, these will expand to be a quartic function of the fermion operators. Problems then arises as it is not trivial to diagonalise a quartic hamiltonian. The genial intuition we were reffering to played by Haldane [160] was to map the problem through a change of the basis into the bosonic operator world, where the interaction is reduced to a quadratic form, hence a closed-form solution is attainable. This process will be briefly outlined in the next section when I will derive the tunnelling density of states in case for a fermion tunneling thorugh a Luttinger liquid.

2.5.3 Derivation of the Tunneling Density of States

We will sketch the derivation of the tunneling density of states (TDOS) for a Luttinger liquid briefly here. The derivation follows reference [167–169]. We will use the second quantization formalism, in which both the wavefunctions and the operators are expressed as operators (more generally, fields) acting in the Fock space. This formalism greatly simplifies the description of multi-electron interacting systems in situations where single-particle interactions fail to capture all the features. The short and medium correlations between different parts of the system, well captured by the Green's function (propagator) between operators associated with a certain observable between different time frames, are central to this formalism.

The tunneling density of states may be written as

$$\rho_{\rm dos}(\omega) = \int_{-\infty}^{\infty} \frac{dt}{2\pi} e^{i\omega t} \left\langle \psi_e(t)\psi_e^\dagger(0) \right\rangle, \tag{2.27}$$

where $\psi_e(t)$ is the time-dependent fermionic operator (field), ω is an energy scale, and the brackets indicate the propagator (Green's function) as instroduced in Sec. 2.5. The mapping between fermionic and bosonic operators takes place through bosonization [161], which allows us to write bare fermionic operators in terms of bosonic fields. We then may write the bosonized form of the fermionic operator as

$$\psi_e = \psi_L + \psi_R,$$

where $\psi_{L/R}$ are the left/right fermionic operators for left-moving and right-moving carriers*, which may be written in terms of a bosonic operator $e^{-i\varphi_{L/R}}$ normalized

*This argument, as will be explained later, assumes that chirality symmetry is respected. As a result, applying a strong magnetic field that may cause chiral-symmetry breaking may compel this entire argument to be adjusted. See [170].

by the inter-particle spacing a when particles are confined along a 1D chain $\psi_{L/R} = e^{-i\varphi_{L/R}}/\sqrt{a}$.

In order to diagonalise the Luttinger liquid Hamiltonian, it may be convenient to introduce the following fields

$$\phi_\pm = \frac{1}{\sqrt{2}} g^{\mp\frac{1}{2}} \left(\varphi_L \mp \varphi_R\right)$$

where $g > 0$ (for free electrons $g = 1$) is a dimensionless parameter which quantifies the electron–electron interactions in a 1D quantum wire. These fields have also the peculiarity to be canonically conjugated to the displacement field $\theta(x)$ which derivative models the electron density fluctuations:

$$\delta n(x) = n(x) - n_0 == \frac{\partial_x \theta(x)}{\pi}, \tag{2.28}$$

where n_0 is the average density. This expression is obtained by considering that if $\theta(a) - \theta(0) = -\pi$, then only one extra electron is between two particles separated by a distance a.

The two fermionic fields thus reads

$$\psi_L = \frac{1}{\sqrt{a}} e^{-\frac{i}{\sqrt{2}}\left(\frac{1}{\sqrt{g}}\phi_- + \sqrt{g}\phi_+\right)},$$
$$\psi_R = \frac{1}{\sqrt{a}} e^{-\frac{i}{\sqrt{2}}\left(\frac{1}{\sqrt{g}}\phi_- - \sqrt{g}\phi_+\right)},$$

while the fermionic propagator may be written as

$$\begin{aligned}\left\langle \psi_e(t)\psi_e^+(0)\right\rangle &= \langle \psi_L(t)\psi_L(0)\rangle + \langle \psi_R(t)\psi_R(0)\rangle \\ &= \langle \psi_L(t)\psi_L(0)\rangle + \text{h.c.}\end{aligned}$$

where h.c. represents the Hermitian conjugate terms in the propagator formula. Cross terms such as $\left\langle \psi_L^\dagger \psi_R \right\rangle$ cancel each other since chirality for this kind of ribbon is conserved. For the remaining terms the sum can be opened such as

$$\begin{aligned}\left\langle \psi_L^\dagger(t)\psi_L(0)\right\rangle &= \frac{1}{a}\left\langle e^{\frac{i}{\sqrt{2}}(\frac{1}{\sqrt{g}}\phi_-(t)\sqrt{g}\phi_+(t))} e^{\frac{i}{\sqrt{2}}(\frac{1}{\sqrt{g}}\phi_-(0)\sqrt{g}\phi_+(0))}\right\rangle \\ &= \frac{1}{a}\left\langle e^{\frac{i}{\sqrt{2g}}\phi_-(t)} e^{\frac{-i}{\sqrt{2g}}\phi_-(0)}\right\rangle \times \left\langle e^{\frac{i}{\sqrt{2g}}\phi_+(t)} e^{\frac{-i}{\sqrt{2g}}\phi_+(0)}\right\rangle\end{aligned}$$

By using the Baker–Campbell–Hausdorff identity, which is valid for exponential operators and is obeyed by free bosonic fields

$$\left\langle e^{B_1} e^{B_2} \right\rangle = e^{\frac{1}{2}} \left\langle B_1^2 + 2B_1 B_2 + B_2^2 \right\rangle$$

and time-translational symmetry, we obtain

$$\begin{aligned}\left\langle e^{\frac{i}{\sqrt{2g}}\phi_-(t)} e^{-\frac{i}{\sqrt{2g}}\phi_-(0)} \right\rangle &= e^{\frac{1}{2g}\langle \phi_-(t)\phi_-(0) - \phi_-(0)\phi_-(0)\rangle} \\ &= \left(1 + \frac{i v_F t}{a}\right)^{-\frac{1}{2g}}\end{aligned}$$

where we introduce the Fermi velocity v_F and used the relation

$$\langle \phi_-(t)\phi_-(0)\rangle = -\ln\left(1 + \frac{i v_F t}{a}\right) \tag{2.29}$$

Similarly

$$\left\langle e^{i\sqrt{\frac{g}{2}}\phi_+(t)} e^{-i\sqrt{\frac{g}{2}}\phi_+(0)} \right\rangle = (1 + i v_F t/a)^{-\frac{g}{2}} \tag{2.30}$$

and by combining the two terms, the fermionic propagator reads:

$$\langle \psi_e(t)\psi_e(0)\rangle = \frac{1}{a}(1 + \frac{i v_F t}{a})^{\frac{1}{2}(g+g^{-1})} \tag{2.31}$$

Incorporating this expression into the TDOS yields:

$$\begin{aligned}\rho_{\text{dos}}(\omega) &= \int_{-\infty}^{\infty} \frac{dt}{2\pi} e^{i\omega t} \left\langle \psi_e(t)\psi_e^\dagger(0) \right\rangle \\ &= \frac{1}{2\pi a} \int_{-\infty}^{\infty} dt e^{i\omega t} \left(1 + \frac{i v_F t}{a}\right)^{-\frac{1}{2}\left(g+g^{-1}\right)}\end{aligned}$$

Since we are interested in the low-energy behavior of the TDOS, we look at the limit for which $\omega \to 0$, which in the time-domain corresponds to look at the behavior for $t \to \infty$:

$$\begin{aligned}\rho_{\text{dos}}(\omega) &\sim \int_{-\infty}^{\infty} dt \frac{e^{i\omega t}}{t^{1/2(g+g^{-1})}} \\ &\sim \omega^{\frac{1}{2}(g+g^{-1})-1} = \omega^\alpha\end{aligned}$$

As detailed very clearly in [169], for a nanotube the exponent is expected to be appreciable $\alpha \simeq 0.4$. In MGNRs, the valley and the spin degree of freedoms amount for a total four-fold degeneracy of the band structure, leading to an overall exponent for tunneling into the end of a MGNR of $\alpha \simeq \frac{1}{4}(g^{-1} - 1)$ [171, 172].

3

Experimental Details: Measurement Protocols and Device Fabrication

Contents

3.1 Introduction

In this thesis, a substantial amount of effort was devoted to the investigation of graphene-based single-electron transistors (SETs) when bottom-up graphene nanoribbons and extended delocalized molecules connect the two leads. To do this and remain within the coherent transport regime (Ch. 2), electrodes separated by only a few nanometres are necessary, as well as cryogenic temperature measurements.

In this chapter, I will describe the room temperature setup we used to fabricate graphene tunnell junctions (GTJs), the development of a new feedback electroburning protocol that increases the chances of having quantum-dot-free GTJs, a few ideas for improving the deposition process, and a brief overview of the cryogenic setup and the electronic apparatus for performing direct current (DC) measurements.

3.2 Room-Temperature Transport Setup

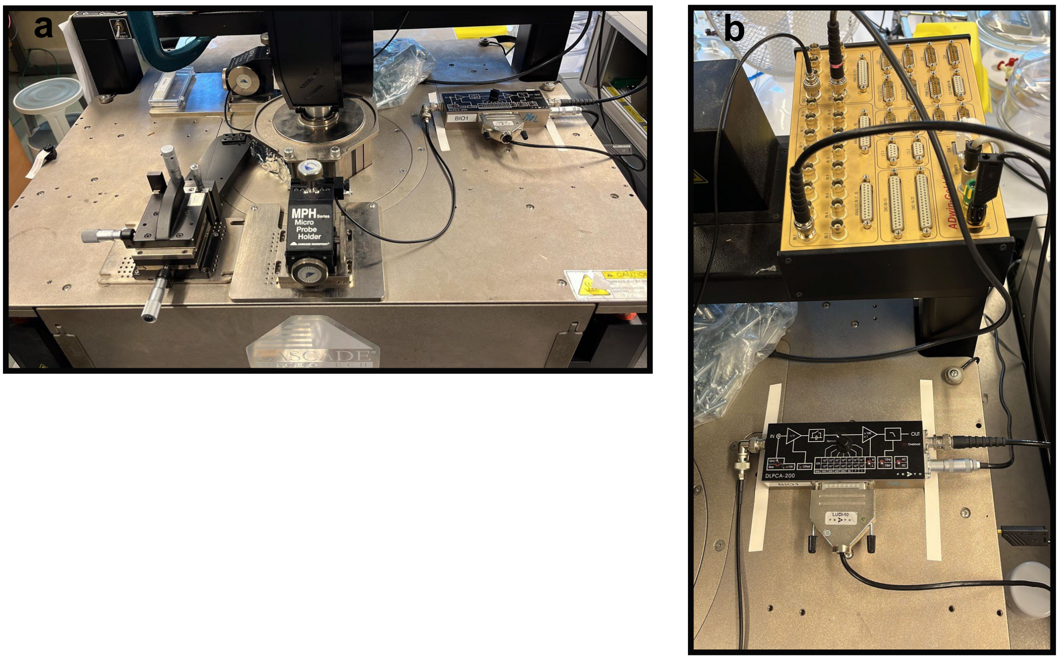

Figure 3.1: a, Probe Station setup. In **b,** the ADwin-Gold II, current amplifier and respective connection for a standard 3T measurement.

Electrical measurements at room temperature are performed on a fully automated Cascade Summit™ 12000 M probe station, fitting three micro-manipulators (Fig. 3.1a). To ensure good electrical contacts, Copper beryllium probes by Lambda photometrics (tip diameter = 25 µm, length = 30 mm) were used for contacting Au (source/drain) pads, Tungesten (tip diameter = 10 µm, length = 30 mm) for HfO_2 (local) pads.

Typically, currents flowing through a quantum device are within the pA − nA range. To detect signals with enough accuracy, a Femto DLPCA-200 low-noise current amplifier with adjustable gain was used to convert currents to voltages (Fig. 3.1b). The bandwidth of the amplifier's filters varies based on the gain settings and operates solely on the pure analogue signal: the internal low-pass-filter (LPF) is approximately 10 Hz to eliminate wide band noise for low-frequency applications, and at the highest trans-impedance gain settable of $1 \times 10^{11}\,\mathrm{V\,A^{-1}}$ guarantees a 3 dB bandwidth of 1.1 kHz. If the gain is not set correctly, the amplifier will overload, and block the passage of more current.

Through an Adwin-Gold II (Fig. 3.1b), voltages are applied and read. The system is equipped with 16 analogue inputs, 2 analogue outputs, 32 digital I/O, and a maximum output voltage range of $\pm 10\,\mathrm{V}$ with a 16-bit resolution. During the feedback electroburning programme, a timer with a resolution of 3.3 ns is necessary for ramping back the voltage at the beginning of each burning cycle. The incoming data stream was sampled at a constant rate of $300\,\mathrm{MHz\,s^{-1}}$.

3.3 Sample Fabrication

An ideal fabrication recipe is one in which small changes in the fabrication process have no substantial effect on the fabrication outcome. Metal pads (Cr 5 nm/ Au 65 nm) were fabricated via standard lithography means (Fig. 3.2). Graphene leads, on the other hand, were fabricated through a modified wet transfer [173] recipe here below detailed.

1. A $1 \times 1\,\mathrm{cm^2}$ piece of graphene is manually cut from a 6 inch sheet of chemical vapour deposition (CVD) graphene, purchased from Graphenea, which is backed with copper foil and coated with a thin protective layer of PMMA on top. The layer thickness we found to work best for our application (i.e. less level of graphene surface contamination) was 300 nm.

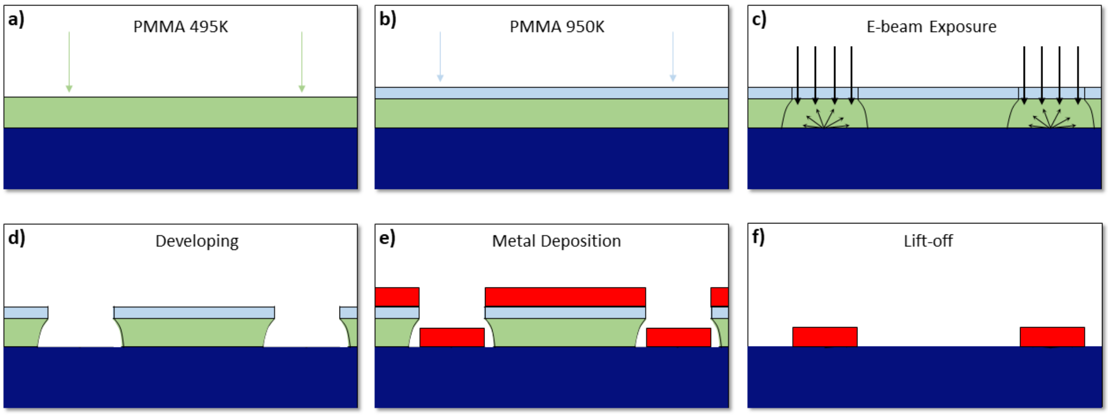

Figure 3.2: Schematic of Metal Deposition. **a)** The first photoresist covers the SiO_2 substrate. **b)** A second layer of photoresist is applied. **c)** Exposure to EBL with secondary electron scattering from the substrate. **d)** Development of the resist, resulting in an undercut. **e)** Metal deposition using a thermal evaporator. **f)** Lift-off aided by the undercut, with the final metal structure remaining on the substrate.

2. The graphene squares are placed into an oxygen plasma cleaner copper side up and etched at 50% power for 60 s with 25 sccm of oxygen to remove residual backside graphene.

3. A 0.4 mol solution of ammonium persulfate (APS) in deionised water is prepared in a clean beaker and sonicated to remove any air bubbles.

4. The graphene pieces are floated on top of the APS solution copper side down and left overnight for the copper to dissolve.

5. $1 \times 1\,\text{cm}^2$ chips of a pre-patterned doped silicon wafer with a 300 nm layer of thermal oxide are cleaned by sonicating in acetone, washing in IPA and drying under nitrogen. The chips are patterned with an array of 851 electrode pairs, formed of 60 nm Au evaporated on top of a 5 nm -thick adhesion layer made by Cr. The electrode pairs are separated by 5 µm at the narrowest point, between which the graphene will be patterned.

6. The etched graphene pieces are transferred three times to clean beakers of deionised water using a clean glass slide to remove residual APS.

7. Graphene is transferred from the water directly onto the pre-patterned chips, using an IPA dip to remove any trapped air bubbles or smooth small creases, and is left to air dry for several hours.

8. Chips are baked at $T = 180\,^{\circ}\mathrm{C}$ for 1 hour to re-flow the protective PMMA layer and remove residual solvents.

9. The chips are submerged in acetone and heated to $180\,^{\circ}\mathrm{C}$ for at least 2 hours to remove the protective PMMA. The chips are then washed in fresh acetone and IPA and dried with nitrogen.

One of the recipe's biggest flaws was the unreliability of graphene wet transfer. Bubbles trapped beneath the water meniscus frequently cause ripples on the graphene surface, compromising the electroburning process [174]. In essence, ripples increase electrical current densities, which leads to increased Joule heating. This frequently resulted in abrupt current drops during the electroburning cyle, and making the GTJs unsuitable for further transport experiment.

To minimise these situations, we later opted to outsource graphene transfers at the wafer size to Graphenea. This enhanced graphene quality and electroburning yields significantly.

3.3.1 Graphene Patterning

Once the Si/SiO_2 substrates were prepared, graphene was deposited onto them following the procedure detailed below.

1. Each graphene coated chip is loaded into a spin coater and $100\,\mu\mathrm{L}$ of Allresist AR-N 7500.18 negative resist is deposited onto the bare graphene.

2. The chip is spun at 5000 rpm for $60\,\mathrm{s}$ in order to achieve a resist thickness of $\simeq 3.5\,\mu\mathrm{m}$.

3. The coated chip is baked at $85\,^{\circ}\mathrm{C}$ for $60\,\mathrm{s}$ to cure the resist.

4. The chip is loaded into an electron-beam lithography (EBL) where constrictions with a nominal width of approximately 100 nm are patterned using a beam current of 500 pA and a dose of 300 µC cm^{-2}.

5. The development step is carried out in DuPont MF-CD-26 developer for 45 s, removing the unexposed resist, before being washed in deionised water and dried under nitrogen.

6. The chips are loaded into an oxygen plasma cleaner and etched at 50% power, 25 sccm of oxygen for 60 s to remove the exposed graphene.

7. The chips are then submerged in Microresist MR-REM 660 at 50 °C for 2 hours to remove the remaining exposed negative resist.

8. The chips are washed in acetone and IPA and dried using nitrogen.

3.4 Graphene Tunnel Junctions

Due to the sp^2 covalent network of bonds between carbon atoms, electroburning does not occur through the migration of single atoms, as it does in gold electromigrated junctions, but rather through more dramatic chemical changes in which the final structure may be significantly different from the initial structure [175, 176]. In contrast to the pristine nanogaps that can be obtained by using metallic nanogaps, control over and knowledge of the final atomic structure of graphene nanogaps are limited, and the relationship between this structure and the electrical transport capabilities of the device is largely unknown.

GTJs were fabricated using by the feedback-controlled electroburning protocol detailed in literature [177, 178]. In a typical electroburning experiment, the voltage across the graphene constriction is ramped up at a rate of 5–8 V s^{-1} to a maximum of 10 V while the current is continuously sampled by the ADWin. Possible inflection points in the first few electroburning traces indicating negative differential conductance might be due to thermal cleaning of the graphene surface from residual litography contaminations [178].

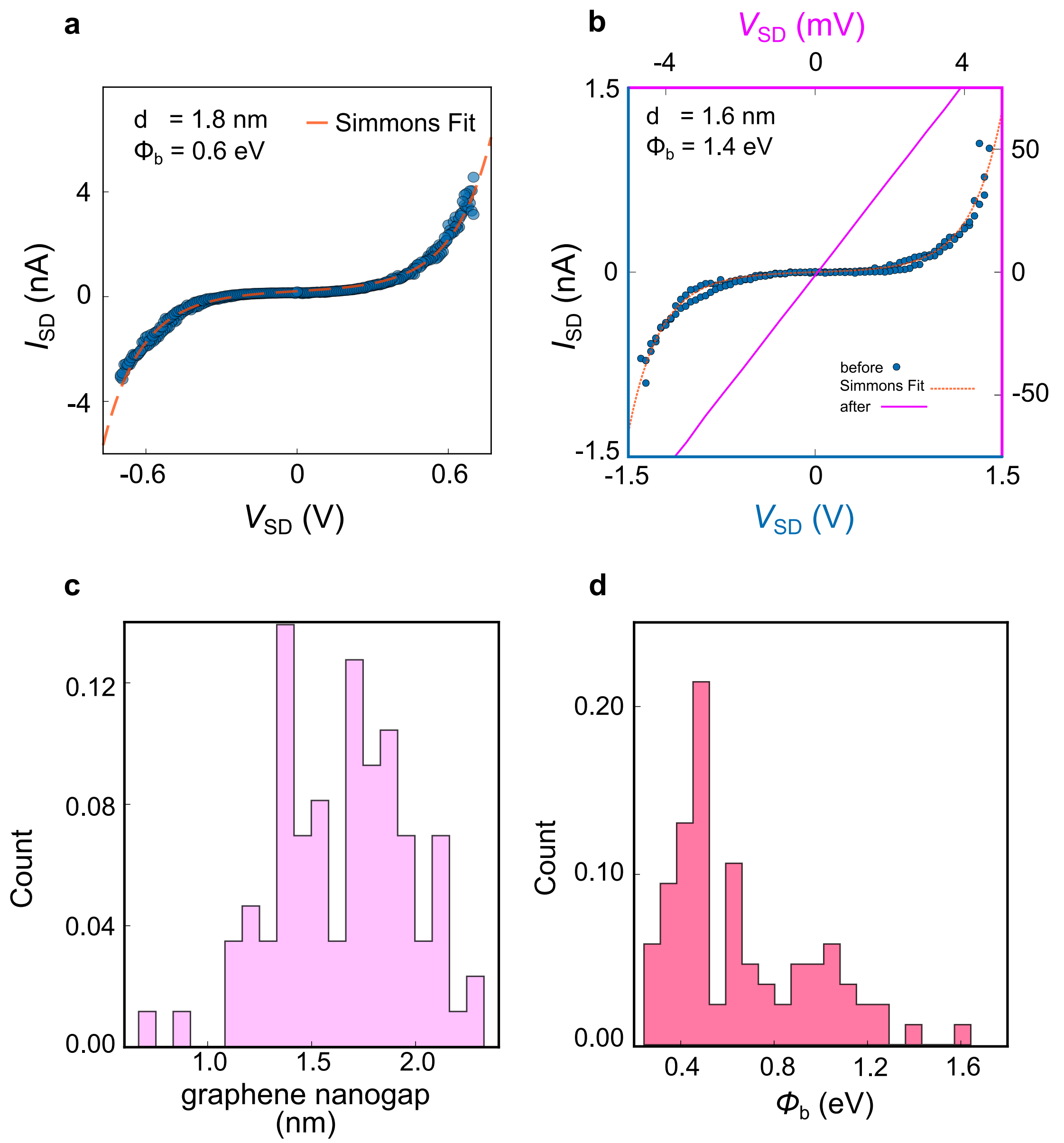

Figure 3.3: **a,** Typical GTJ measured at zero gate at the end of the electroburning. **b,** Source drain current I_{SD} pre (blue) and post (purple) depositing a solution of GNRs. **c,** Histrogram of the nanogap size resulting by fitting I_{SD} with the Simmons model as explained in the main text. **d,** A Poisson distribution-like histogram of the energy barrier fitting parameter.

When the current decreases as a result of removal of carbon atoms from the graphene lattice due to Joule heating, the feedback protocol responds by ramping the voltage back to 0 V at a rate of about 2500–5000 V s^{-1}. The process is repeated until the junction's resistance exceeds a specified threshold, which has been modified to optimise the process. Typical threshold values ranged between 0.5–2.4 GΩ, yielding a clean-junction percentage of roughly 18–24 %.

Fitting the $I - V_{\mathrm{SD}}$ trace of a device (Fig. 3.3a,b) to the Simmons model with an asymmetric barrier

$$I_{\mathrm{SD}}(V_{\mathrm{SD}}) = I_0 + \frac{Ae}{(2\pi h d^2)}\Bigg[(\Phi_{\mathrm{b}} - eV_{\mathrm{SD}}/2)\exp\left(\frac{-4\pi d}{h}\sqrt{2m\left(\Phi_{\mathrm{b}} - eV_{\mathrm{SD}}/2\right)}\right) - (\Phi_{\mathrm{b}} + eV_{\mathrm{SD}}/2)\exp\left(\frac{-4\pi d}{h}\sqrt{2m\left(\Phi_{b} + eV_{\mathrm{SD}}/2\right)}\right)\Bigg]$$

allows us to retrieve several fitting parameters. I_0 is an offset current accounting for possible shifts along the y-axis, A is the contact area of the two electrodes (cross-sectional area), d the gap size and Φ_{b} is the potential barrier expressed within a WKB approximation for transmission probability of an electron to tunnel through a an insulating barrier.

Fig. 3.3c,d shows the histograms for gap widths, which has mean gap size of approximately $\bar{d} = (1.7 \pm 0.3)$ nm, and for the energy barrier of $\overline{\Phi_{\mathrm{b}}} = (0.7 \pm 0.4)$ eV.

3.4.1 Gate-controlled Electroburning

When molecules are deposited on graphene leads, the tunnel coupling resulting from the anchoring to the honeycomb lattice frequently results in systems with a weak coupling regime (see Ch. 2). This indicates that Coulomb blockade characteristics can (and typically are) seen at gate voltages other than zero. This is problematic because these characteristics, which originate from contaminations formed during electroburning, can frequently obscure molecular features. To mitigate this problem, a novel electroburning procedure was created.

The computational flowchart that was subsequently implemented within the Python measurement framework QTlab [179] is seen in Fig.3.4. Essentially, the protocol runs a first electroburning cycle and then determines if the probability distribution of the source-drain current values recorded via a gate trace at a fixed bias differs from a Gaussian distribution.

The null-hypothesis is evaluated using the Shapiro–Wilk test, and is rejected if the p-value is $p \leq 0.05$: in this case, there is a 95% chance that the source–drain current values do not follow a normal distribution, hence there must be a residual quantum dot at finite gate voltage. Visual inspection typically supports this conclusion by observing the progressive disappearing of current peaks in a gate trace (Fig. 3.4). Subsequently, the gate is automatically set to the maximum current value in the gate trace, and a second electroburning cycle begins. The loop is performed a predefined number of times, which is often set to 3. By leveraging this experimental GTJ fabrication protocol the percentage of clean junctions increased up to 52%–66%.

Despite the fact that this procedure has increased the overall yield of clean junctions, several refinements are still possible. A future improvements of this approach should accommodate for situations in which significant signal spikes induced by frequency pickup could be identified as the maximum current point even though there is no coulomb diamond present. For instance, to distinguish the current steps from the background noise, an iterative median filter might be applied to the data to filter away transient spikes in the conductance while keeping even modest conductance steps caused by charging effects. After normalising the data, it is possible to separate the relevant steps from the background noise by evaluating the null-hypothesis for a normality test and counting only if the heights of the related conductance steps exceed a predetermined threshold.

3.5 Contacting Graphene Nanoribbons

The problem of connecting graphene nanoribbons or any molecular adduct to a pair of electrodes is one of the greatest challenges to the development of reliable and robust electronics. As described in Ch. 1, when graphene is contacted with metals, the large difference between the two work functions would produce finite Schottky barriers and the metal–graphene interface.

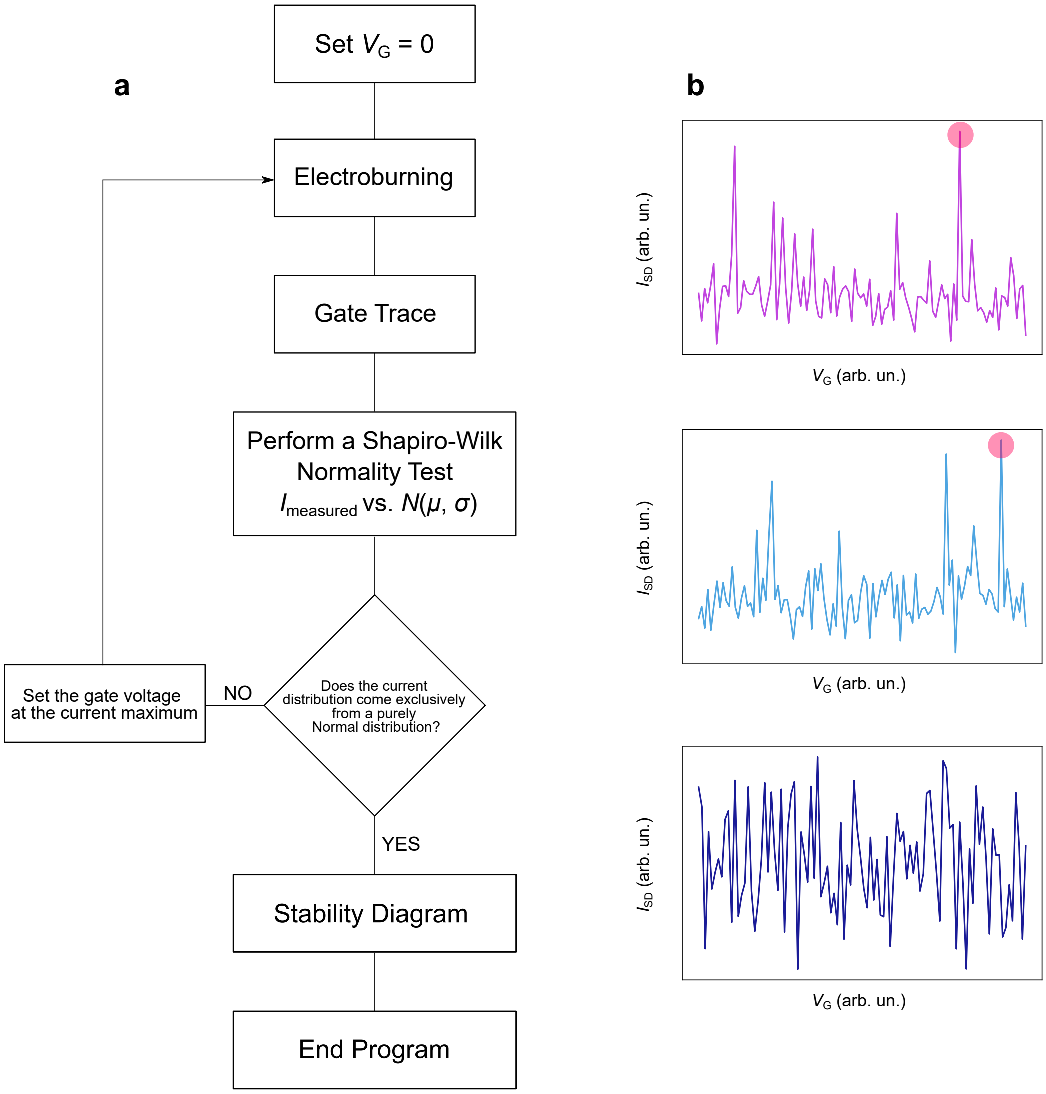

Figure 3.4: Gate-controlled feedback electroburning routine. **a,** Block diagram for the program as implemented. **b,** Gate trace examples as they are extracted at various stages of the routine. Usually a 0.4 V bias is applied during during the gate trace scan. Top graph reports an average gate trace taken at the end of a first electroburning cycle. The program then reads the maximum of the current (red circle), set the gate output channel of the ADwin to that corresponding potential, and starts a second electroburning protocol. When the distribution of the current data passes the Shapiro–Wilk statistics test (bottom graph), the null-hypothesis of the test is accepted (H_0 = the population of the current data is normally distributed), and the program performs a final stability diagram and save all the data in suitable format.

Liquid-processable GNRs offer a step forward in terms of structural integrity and edge shape. One of the major hurdles to the usage of ribbons in solutions is the positioning of such structures on top of a device. When a drop of solution, even if poorly concentrated, is deposited on a chip, a number of competing events occur.

To begin, while still suspended in the solvent, GNRs will be subjected to two opposing forces. The solvent molecules will try to maximise the number of electrostatic contacts, and hence the number of solvent molecules adsorbed on ribbon surface, thus resulting in a flat strand, but intramolecular forces will bend up the nanoribbon. Furthermore, when a ribbon tries to fall on top of the electrode pairs, two conflicting mechanisms are at work. The anchoring of the ribbon with the leads will be preferred from a thermodynamic aspect if the free energy released after the creation of the anchoring bonds is larger than the energy required to keep the ribbon in solution. Second, if the solvent evaporation rate is too quick, the thermodynamic regime will not be achieved, and it will be more likely that molecular aggregates will be formed on top of the device surface, resulting in poor device yields (see al Sec. 3.6).

The problem of a graphene nanoribbon landing on a pair of electrodes separated by a gap is analogous to a well-known problem is geometric probability called the Buffon's needle. The Buffon's needle problem, first proposed in the eighteenth century, estimates the likelihood that by randomly tossing a needle of length l this will touch at least one line of a group of parallel lines separated by a distance $d > l$. Even if no interactions between the needle and the grid are assumed, the problem captures the core of our experiment. The probability that the needle intersects at least one line can be computed analytically in the limit where $\lambda = l/d > 1$ [180]

$$P(\lambda) = \begin{cases} \frac{2\lambda}{\pi} & \lambda \leq 1 \\ \frac{2}{\pi}\Big(\lambda - \sqrt{\lambda^2 - 1} + \sec^{-1}(\lambda)\Big) & \lambda > 1 \end{cases} \tag{3.1}$$

When λ is smaller than zero (Fig. 3.5), the coverage probability grows linearly with the length of the ribbon. This conclusion can be simply explained by observing that for a fixed grid-spacing distance, using increasingly longer ribbons increases our chances of successfully crossing a graphene nanogap. However, expanding the ribbon length forever is not recommended because the crossing probability drops

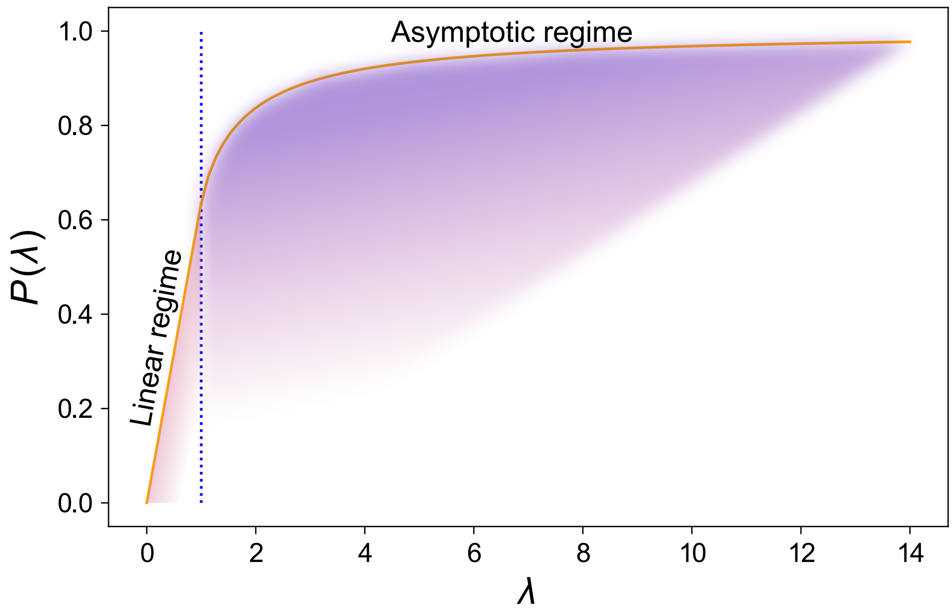

Figure 3.5: Probability distribution as a function of the parameter λ for a needle tossed onto a grid of equispaced parallel lines. When the needle is short ($\lambda < 1$), the coverage probability increases linearly with the length. When the needle becomes larger than the gap ($\lambda \geq 1$), the probability increases asymptotically to one and is independent of the parameter λ.

below one for ribbons much longer than the grid spacing. This would result in a device surface covered in ribbons arranged in random directions, with many ribbons covering the same surface area, and increasing the odds of obtaining ribbon adducts and aggregates. This regime may be sufficient for the fabrication of field-effect transistors, but it should be avoided if the goal is to produce single-electron transistors with only one ribbon bridging the nanogap. These straightforward and intuitive ideas have recently been confirmed by atomic force microscope (AFM) and electrical transport measurements [181].

3.6 Deposition Process and Related Phenomena

One of the key point of a device fabrication are the complex interactions establishing as a consequence of various surface-related phenomena which occur when a molecular adduct has to be transferred onto specific substrates [182, 183]. The rich and wide physics GNRs have displayed on substrates where are directly growth remains largely un-matched when ribbons are transferred onto electronic devices; therefore, I argue that a more detailed look at the deposition picture is essential for improving device yield and reproducibility. In order to fulfill this purpose, I qualitatevely studied the effect of different depositon techniques when a solution of liquid-processable nanoribbons was dropcasted on top of the substrate. By substrate here I refer to the a chip where gold and graphene pads have been shaped.

GNRs were first diluted in an organic apolar solvent before being dropcasted onto the substrate. Apolar solvents have been a valid choice in order to prevent unwanted charge-doping effects of graphene sheets. I have preferably used toluene, choloroform and tetrahydrofuran (THF) to this purpose: this were largely based on suggestions from our collaborators who synthesised the GNRs I have used and by Dr Xuelin Yao. I have also attempted to use higher boiling points solvents such as 1-phenyloctane. The logic behind this choice is that the depositon process is largely dependend on the interactions established between the ribbon and the different surfaces on to of the chip.

In order to favour the $\pi - \pi^*$ interactions between the graphene leads and a single GNR strand, a thermodynamic equilibrium between the amount of substance i absorbed on the electrode per unit area, Γ_i, and in the bulk (the ribbon in the solvent) should be favoured compared to kinetic-related processes such as mass transfer by migration and diffusion.

An estimate of these interactions is provided by first-principle calculations where different anchoring groups such as pyrene and hexabenzocoronene were tested [178]. Approximately, the interaction between a graphene sheet and a pyrene molecule was calculated to be 26.3 meV/atom. The large-width limit for the binding interaction between two graphene sheet reports ranges within this estimate: 10–30 meV/atom for a stacking of type AA with an interlayer distance of 3.3 Å, and 18–23 meV/atom for AB-type stacking [184].

The deposition strategy we adopted to fabricate GNRFETs was inherited from the CNT literature and is known as *dry-cycling* incubation [185]. The

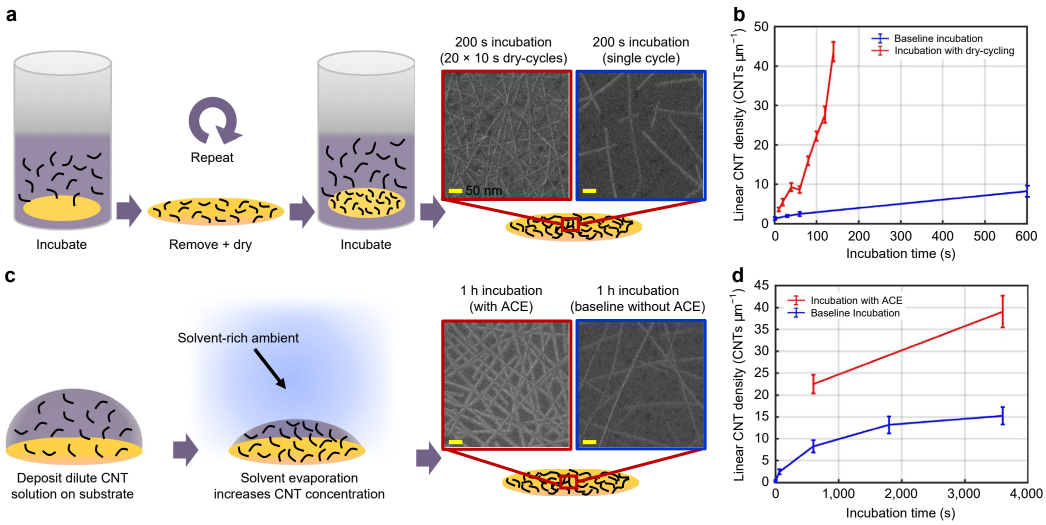

Figure 3.6: a, Cartoon of the dry-cycling process. After a short period of incubation (10 s), the substrate is removed, dried, and reintroduced into the CNT solution. This process is repeated for at least 10 times. **b,** Comparison of deposition rates for baseline incubation (single-cycle deposition) and dry-cycling incubation. This corresponds to a 1.5-fold increase in linear density and a $> 1,000$-fold increase in speed in comparison to the baseline incubation. Error bars indicate the 97.5% confidence interval. **c,** Cartoon of artificial concentrated evaporation (ACE). A low-concentrated GNR solution is deposited onto the substrate which is incubated in a solvent-rich environment. As the solvent evaporates, GNRs solution increases. **d,** Comparison of the linear density attained during ACE incubation versus the initial incubation. After $\simeq$ 1h, ACE achieves a 2.5-fold increase in density relative to the initial incubation at the same CNT stock concentration. Error bars depict the 97.5% confidence interval. Adapted from [185].

technique leverages the Frumkin isotherm to limit the GNR desorption in favour of a faster deposition rate. The substrate is submerged in a GNR solution and incubated for 10 s seconds, leveraging the fast, linear deposition regime outlined in Fig. 3.5. After removing it from the solution, it is washed with IPA and gently blowdry with nitrogen.

This *single-cycle* incubation is repeated several times. Fig. 3.6b shows the CNT linear density together with control samples, and put in evidence the higher surface coverage ($\simeq$ 45 CNT/μm after 150 s).

In addition to slowing the desorption rate, increasing the rate of adsorption is complementary to the dry-cycling method. In order to achieve this, and according to the chemical equilibrium principle, evaporation of the solvent would be slowed down when this evaporates in a gas-phase-rich atmosphere. With this method, also known as ACE (Fig. 3.6c,d), the solvent evaporates graduallty from the solution over time, increasing the concentration of GNRs above the initial concentration. Importantly, the rate of solvent evaporation is crucial for uniformity because sliding during rapid droplet evaporation generates *coffee ring* effect, resulting in non-uniform deposition throughout the substrate [186].

In chapters 5 and 6, transport spectroscopy at milliKelvin temperatures reveals that an improved GNRs solubility is paramount for studying their fundamental physiochemical properties when integrated in a working electronic device. This enhancement has to be sought in the bulky (maleimide) functionalisation introduced thorough a Diels–Alder reaction. This may be schematically written as

$$\mathrm{MGNR} - \mathrm{Anthracene} + \mathrm{Maleimide} \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} \mathrm{MGNR} - \mathrm{Maleimide}, \tag{3.2}$$

This reaction is reversible, hence it follows the Le Chatelier's principle of equilibrium. When the temperature is increased, the entropic contribution dominates the reaction free energy, which favours the reversal of the equilibrium direction. Even if a faster solvent evaporation is desired after deposition on the device substrate, heating should be avoided. Particularly, the reverse reaction becomes favoured either by heating the solution to close to 100 °C, in which case the reaction rate will make the reaction quick, or by maintaining the temperature at approximately 80 °C for an extended period of time.

As a result, we had to employ solvents with lower boiling points for an optimal deposition. We found THF and toluene to perform significantly better in this regard than 1-phenyloctane or 1,3,5-trichlorobenzene.

3.7 Cryogenic-Temperature DC Transport Setup

Transport experiments at milliKelvin temperatures were performed in a Triton 200 Oxford Instrument dilution refrigerator equipped with a 3D vector rotate magnet (VRM). The schematic of the instrumentation is outlined in Fig. 3.7. The operation principles can be found in the dedicated manual provided by the Oxford Instruments.